



Indian Institute of Technology Bombay
Continuing Education &
Quality Improvement Programmes (CE & QIP)

CERTIFICATE

This is to certify that

G. Sanjeevkumar

has participated in the AICTE Sponsored one-week QIP Short Term Course on

Theory of Plasticity and its Application
conducted by IIT Bombay during November 25 - 29, 2019

Preeti Rao

Prof. Preeti Rao

Professor-In-Charge (CE & QIP)



Please Scan QR Code OR type code in <http://www.cep.iitb.ac.in/cep/verify/>
Q5-1920-10

Prashant Date

Prof. Prashant Date

Course Coordinator



Course Contents

- Stress and strain analysis
- Basic constitutive equations
- Maximum friction law in plasticity
- Asymptotic singular representation of rigid perfectly plastic solutions near maximum friction surfaces and limit analysis
- Experiment on generating a layer with drastically modified microstructure near friction surface
- Numerical method for calculating the strain rate intensity factor for rigid perfectly plastic material and double shearing model
- Ideal flow condition
- Tutorial – I, II, III, IV, V
- General theory of stationary planar ideal flow for rigid perfectly plastic material
- General theory of non-stationary planar ideal flow for rigid perfectly plastic material
- General theory of stationary axisymmetric ideal flow for Tresca material
- Practical applications of ideal flows
- Ductile fracture in ideal flows
- General theory of stationary and non-stationary planar ideal flow for the double shearing model